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The vehicle routing problem in field logistics part I

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The vehicle routing problem (VRP) has been characterised as one of the great success stories of operational research, providing and facilitating, for over fifty years, optimal planning solutions for vehicle fleets in a large number of real-life applications. In its simplest form, a VRP can be described as the problem of determining least-cost routes from one depot to a set of geographically dispersed "customers" (such as cities, stores, warehouses, and schools). VRP constitutes one of the most challenging combinatorial optimisation problems. A large number of different approaches have been developed over the years and a number of software packages are available on the market. In the agriculture, the VRP has only very recently been applied to the planning and execution of in-field operations, despite the fact that almost all agricultural field operations (AFO) inherently involve the motion of vehicles.

In this paper, a dedicated classification of AFO is devised and tailored to a conceptual application of the VRP within the domain of agricultural field logistics. The concept regards the operations of primary agricultural machines, with the worked tracks in the field representing the "customers" using VRP methodology. By casting the agricultural fleet management problem in the framework of well-known VRP instances, over half a century of research and implementation experience can be brought to bear on the problem. This will provide for the enhanced management of conventional machinery systems, as well as, the mission and route-planning aspects of the high level control of emerging field robots.

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1. Introduction

A typical vehicle routing problem (VRP) determines the least-cost routes from one depot to a set of geographically dispersed points (such as cities, stores, warehouses, and schools). The routes must be generated in such a way that each point is visited only once by exactly one vehicle with all routes starting and ending at a depot, so that the total demands of all points on one particular route will not exceed the capacity of the vehicle. The problem was first introduced by Dantzig and Ramser (1959) to describe a real-world application concerning the optimum routing of a fleet of gasoline delivery trucks between a bulk terminal and a large number of service stations supplied by the terminal. The next milestone occurred a few years later, when Clarke and Wright (1964) proposed an effective greedy heuristic algorithm that improved the Dantzig-Ramser approach.

Since then, a large body of literature has been produced, including a number of extensive reviews (Desrochers *et al.*, 1990; Toth and Vigo, 2002). Numerous applied variations of the problem in real-world systems have been addressed, including public transport, movement of industrial goods

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Nomenclature

A_i^{index}	the set of nodes corresponding to: (i) index ≡ Ø, the tracks of sub-area i $A_i \subset T, i \in \{1, \ldots, m\}$, (ii) index ≡ [1], the upper headland endings of the tracks represented by A_i and (iii) index ≡ [0], the lower headland endings of the tracks represented by A_i
E	the set of arcs (edges) of the graph $G : E = N \times N$
C_{index}	the capacity of a mobile unit: index ≡ $i \in V$, in general the capacity of a vehicle as in VRP, index ≡ $p_k, k \in P$, capacity of a PU, index ≡ $s_k, k \in S$, capacity of a SU
C_{ij}	the cost associated with each arc $(i, j) \in E$ representing the transit cost from i to j
d_i	the nonnegative demand that is associated with each customer $i \in N$
F	the set of the facility units, each referenced as f_k, $k \in F$
G	the graph $G = \{N, E\}$
$\tilde{h}(i,j)$	the time that a machine needs in order to execute a headland turning from the end of the track represented by node i to the start of the track represented by node j
N	the set of nodes in the VRP graph
P	the set of the primary units, each referenced as p_k, $k \in P$
R_j	the permutation set that includes the ordering of the nodes visited by the vehicle v_j, $j \in V$ according to VRP solution, $R_j = \langle i_1, i_2, \ldots, i_{R_j} \rangle$
S	the set of the service units, each referenced as s_k, $k \in S$
T	the set of the field tracks that cover the entire field area
V	the set of the vehicles available at the depot, each referenced as v_k, $k \in V$
$\gamma(\cdot) : \mathcal{N} \rightarrow \{0, 1\}$	the function that provide as result the parity of an integer: $\gamma(i) = 1$ if the integer i is an even number and $\gamma(i) = 0$ if i is an odd number
$\Delta(i)$	the set of nodes j that are directly reachable from i $(i, j \in N)$
$\Delta^+(i)$	the *forward star* of node $i \in N$, it is defined as the set of vertices $j/(i,j) \in A$ that are directly reached from i
$\Delta^-(i)$	the *backward star* of node $i \in N$, it is defined as the set of vertices $j/(i,j) \in A$ from which i is directly reachable
δ	the operator that provides the connectivity of the graph between the sub-areas

Abbreviations

AFO	agricultural field operation
aVRP	asymmetric vehicle routing problem
CTF	controlled traffic farming
DCVRP	distance constrained vehicle routing problem
DVRP	dynamic vehicle routing problem
IMF	input material flow
FU	facility unit
m-TSP	multiple travelling salesman problem
MFVRP	mixed fleet vehicle routing problem
NMF	neutral material flow
OMF	output material Flow
PU	primary Unit
SU	service Unit
SVRP	stochastic vehicle routing problem
sVRP	symmetric vehicle routing problem
TSP	travelling salesman problem
TSPB	travelling salesman problem with Backhaus
TSPTW	travelling salesman problem with time windows
VRP	vehicle routing problem
VRPB	vehicle routing problem with Backhaus
VRPPD	vehicle routing problem with pick-up and delivering
VRPSD	vehicle routing problem with stochastic demands
VRPTW	vehicle routing problem with time windows

along a supply chain, urban solid waste collection, street cleaning, school bus routing, routing of salespeople, and courier services. A number of software packages are available on the market for use in a number of these applications.

The VPR is a generalisation of the travelling salesman problem (TSP), in which only one person has to visit all the customers. The TSP can be generalised to the case of multiple TSP (m-TSP) in which more than one salesman is involved in servicing the customers; this adds capacity constraints to the case of the classic VRP. Addition of a new set of constraints results in a number of VRP instances. An example would be the constraint of maximum length or duration of a route (distance constrained VRP – DCVRP) or that every customer should be served within a given time window (TSP with time windows – TSPTW and VRP with time windows – VRPTW, for the cases of single- and multiple-vehicle problems, respectively). Furthermore, in some cases the possibility that customers will return some commodities should be considered (VRP with pick-up and delivering- VRPPD), and that all deliveries must be made on each route before any pickups can be made (TSP with Backhaus- TSPB and VRP with Backhaus-VRPB, for the cases of single- and multiple-vehicle problems, respectively). Finally, in practise, requests for service can often arrive sequentially in time, with stochastic inter-arrival times. Moreover, locations and quantities of future demands may be unknown or known only probabilistically, as many routing problems are inherently *dynamic* (DVRP) and *stochastic* (SVRP). Fig. 1 depicts the interconnections between all the previously mentioned generic VRP instances. More recently, attention has been devoted to more complex variants of the VRP (usually called ''rich'' VRPs) that are closer to the practical distribution problems than classic VRP models. These complex variants are characterised by such variables as multiple depots, multiple trips, multiple vehicle types, and loading constraints (Bräysy *et al.*, 2002; Cordeau *et al.*, 2006).

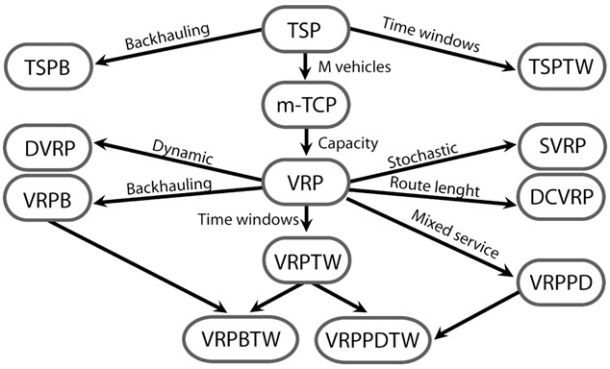


Fig. 1 – Relationships between the basic VRP instances.

VRP constitutes one of the most challenging combinatorial optimisation problems, and a large number of different problem solving approaches have been developed over the years. Among them, exact methods: such as branch-and-bound (Fisher, 1994); branch-and-cut (Lysgaard *et al.*, 2003); heuristic methods (Cordeau *et al.*, 2002); metaheuristic methods: such as simulated annealing (Kirkpatrick *et al.*, 1983), tabu search (Amberg *et al.*, 2000), genetic algorithms (Thangiah, 1995), and ant algorithms (Reimann *et al.*, 2004).

In the field of agriculture, the VRP has been implemented in the scheduling of livestock or the transportation of agricultural products (Sigurd *et al.*, 2004; Oppen and Løkketangen, 2008) as well as in food logistics applications (Tarantilis and Kiranoudis, 2004) analogous to other general commodities logistics applications. Basnet *et al.* (2006) implemented the TSP for the farm-to-farm path determination as part of a proposed method for the scheduling of crop harvesting in multi-farm operations.

In contrast, and despite the fact that almost all agricultural field operations (AFO) inherently involves motion of vehicles, the VRP problem has only very recently been implemented for the planning and execution of in-field operations involving agricultural machinery. Bochtis (2008) showed the potential of the VRP application in cases of single or multiple machinery systems operating in one or multiple geographically dispersed fields. According to the procedure introduced by Bochtis (2008), field coverage is expressed as the traversal of a weighted graph, and the problem of finding optimal traversal sequences is shown to be equivalent to finding the shortest tours in the graph. An implementation the procedure in conventional agricultural machines, with auto-steering systems, was presented in Bochtis and Vougioukas (2008). The experimental results showed that, by using a specific set of algorithmically computed optimal sequences based on the VRP approach, the total non-working distance can be reduced significantly by up to 50%. The same approach has been implemented for the mission planning of an autonomous tractor for area coverage operations such as grass mowing, seeding and spraying (Bochtis *et al.*, 2009) and orchard operations, using the capacitated VRP (Bochtis *et al.*, 2008). Alia *et al.* (2009) proposed a combination of VRP and minimum cost network flow problems in order to determine the optimal covering routes for combine harvesters, as well as feasible positions for grain transfer between the combine harvesters and tractors.

This paper presents conceptual applications of VRP within agricultural fleet management. The planning of a number of field operation cases is cast as instances of VRP, using the abstractive representation of the worked tracks in the field as the ''customers'' in the VRP methodology. Using this abstraction, it is shown that the inherent planning problem in AFO is the problem of the optimal traversal of weighted graphs under a number of operational constraints. Consequently, route and mission planning for agricultural vehicles, for conventional and distributed mobile robot systems, can be formalised as a VRP. As a result, field operation planning could benefit from five decades of research into routing problems.

2. Preliminaries

2.1. VRP notations

The basic form of the VRP is related to a graph $G=\{N,E\}$, where $N=\{0,1,\ldots,n\}$ is the node set and E is the arc set. Node 0 corresponds to the vehicles' depot while the remaining node set $N\backslash\{0\}$ corresponds to the customers. A non-negative cost $c_{ij}$is associated with each arc $(i,j)\in E$ representing the transit cost (expressed in time or distance) from i to j. In the $c_{ij}=c_{ji}$ case, the problem is called symmetric (sVRP); otherwise, it is called asymmetric (aVRP). Each customer, $i\in N/\{0\}$, is associated with a non-negative demand of d_i units (d_0=0). A number of identical vehicles (let $V=\{v_1,v_2,\ldots\}$ denotes the set of them), each with capacity C, is available at the depot.

The VRP consists of determining one route for each vehicle ($|V|$ permutation sets of the form $R_j=\langle i_1,i_2,\ldots,i_{|R_j|}\rangle$), with minimum cost, satisfying the following constraints:

(i) Each route starts and ends at the depot, i.e., $i_1=i_{|R_j|}=0$, and $\{i_2,\ldots,i_{|R_j|-1}\}\subseteq N\setminus\{0\}$
(ii) Each customer is visited by exactly one vehicle, i.e., $\cap_{j=1}^{|V|}R_j=0\wedge\cup_{j=1}^{|V|}R_j=N$
(iii) The total demand of costumers visited in a route does not exceed the vehicle capacity, i.e., $\sum_{h=2}^{|R_j-1|}d_{i_h}\le C$.

Furthermore, for a feasible solution of the problem, it is assumed that $d_i\le C,\forall i\in N/\{0\}$ and[1] $|V|\ge\lceil\sum_{i\in N\setminus 0}d_i/C\rceil$.

2.2. Definitions

In the following, a classification of the AFO, in terms of the machinery system, the operation nature and the facility units' characteristics, is presented. The definitions of terms that will be used in the rest of the paper and the corresponded symbols are also given.

A summary of the classification using the dedicated symbols is given in Table 1.

Let P denote the set of the PU, each referent as $p_k,k\in P$, S denote the set of the SU, each referent as $s_k,k\in S$, and F denote

[1] Symbol $\lceil\rceil$ denotes the ceiling function, while symbol $||$ denotes the cardinality of a set.

<Machines>	<Type>	*Primary Unit* (*PU*). This term refers to a unit (self-propelled machine or tractor-implement combination) that performs the main work task. Examples include a self-propelled harvester, a tractor carrying a sprayer, a cultivator, *etc.* *Service Unit* (*SU*). This term refers to a unit supporting one PU or a team of PUs. Examples include transport waggons in a harvesting operation being used for out-of-the field removal of harvested grains or units supporting application units by carrying the fertiliser for their refilling.
	<Number>	One machine (1). Multiple machines (*m*)
	<Team identity>	This refers to the case of multiple-machines and specifies whether the team is composed of identical machines or different machines in relation to their operational features. *Identical machines* (≡). *Non-identical machines* (≠ **{*kin,cap*}**). The operational specifications that need to be taken into consideration for any VRP variant applied to the AFO. It is the operating width, in the case of a PU, and the quantity of material it can carry, in cases of PU as well as SU. In this paper, the approach will be limited to derivations on the former. Nevertheless, issues regarding diversions on the fixed and operational cost of the machines of a team will be mentioned in the discussion section.
<Operation>	<Material Flow>	*Input material flow* (*IMF*). In these operations, a quantity of a ''commodity'' is transported by the machine and is distributed in the field area (e.g., saeed, spraying solution or fertilizer). *Output material flow* (*OMF*). In these operations, a quantity of a ''commodity'' is transported out of the field area (e.g., harvesting). *Neutral material flow* (*NMF*). In these operations, there is neither material addition nor material removal to/from the field (tillage, cultivation, seedbed preparation, mowing, hay conditioning, hay baling, windrowing, etc.).
	<Demand Characteristics>	This refers to IMF and OMF operations and has to do with the required information regarding the quantity of material that has to be removed from, or distributed in, the field. *Deterministic* (+). This is the case of *a priori* knowledge of demand, for example, in a seeding operation; the saeed quantity per field area unit is predetermined. *Stochastic* (∼). In this case, there is no precise knowledge of demands, but it is expected to follow known probability distributions, for example, the predicted yield distribution. *Unknown* (!). In this case, the actual demand is unknown and there is no probabilistic knowledge of it, for example, the operation of real-time spraying based on sensor data.
<Facilities>	<Number>	One facility unit (1). Multiple facility units (*m*)
	<Type >	*Stationary* (*sta*). For example, a farm depot or a silo. *Mobile* (*mob*). For example, public road transport trucks.

Table 1 – AFO classification.

Machines			Operation		Facility units	
Type	Number	Team identity	Material flow	Demand type	Number	Type
PU	1	≡	IMF	+	1	sta
SU	M	≠ (*kin,cap*)	OMF	∼	M	mob
			NMF	!		

the set of the FU, each referent as $f_k, k \in F$. All machines included in sets *P* or *S* are considered homogenous[2] (of the same type).

[2] It is necessary to distinguish the use of the term ''homogenous'' in VRP and AFO. In the VRP literature, the term refers to vehicles that are identical in terms of their operational specifications, e.g., those having the same capacity, same fixed cost, same distance travelled and costs, same time of use and costs, etc. In the AFO case, the term ''identical'' is used to state the previous, while the term ''homogenous'' will refers to machines of the same operation type, e.g., harvesters, fertilisers, and transport trailers.

3. Casting agricultural field operations into examples of vehicle routing problem

The following concerns field area coverage operations (in arable farming as well as in orchard operations) where PUs operate following a headland fieldwork pattern. In these patterns, a given field is covered by a set of parallel tracks, or trips, which start at one boundary of the field and terminate at the opposite boundary. In Bochtis and Vougioukas (2008), the representation of each field track of a simple-shaped field was presented by one node in an undirected complete graph. The current presented approach constitutes an evolution of the previous that includes complex-shaped fields consisting of sub-areas and multiple-fields, as well as trafficability constraints imposed between field

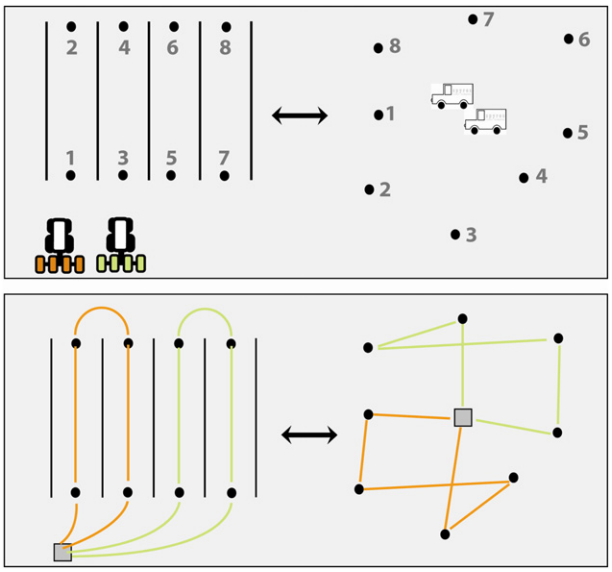


Fig. 2 – Correspondence between agricultural machinery operations and VRP.

areas or field tracks. According to this approach, each field track is represented by two nodes, one for each track end, corresponding to two customers of a VRP (Fig. 2).

3.1. *Graphical representation*

Let T={1,2,3,…} be the arbitrarily ordered set of the field tracks that cover the entire area under study. According to this representation, the set N of nodes of the corresponding VRP is expressed as$N = \{0, 1, 2, 3, .., 2|T|\}$. Suppose that the area at hand is composed by m sub-areas[3], which can be sub-fields of a field, multiple geographical dispersed fields, or can include both categories. Let $A_i \subset T, i \in \{1, \ldots, m\}$ be the set of tracks that area i consists of. Nodes of the same parity, either odd or even, correspond to track ends that belong to the same headland (Fig. 2). It is possible to arbitrarily designate the headland, in which track ends are represented by odd nodes, as the lower headland of the sub-area, while the headland in which track ends are represented by even nodes is designated the upper. Let $A_i^1, A_i^0 \subseteq N$, represent the sets of nodes corresponding to the upper and the lower track endings, of area I, respectively. The routes of a feasible solution should not include connections between two nodes that represent endings of different tracks at opposite headlands. Such a representation of the field provides a non-completed graph (there are missing arcs in the graph). We define[4]

[3] The term "sub-area" describes a set of tracks for which there are not trafficability constraints for a mobile unit travelling between each track through the headland.

[4] There are two sets that are used in the VRPs: the so-called *forward star* of i, $\Delta^+(i)$, which is defined as the set of nodes j :$(i,j) \in E$ that are directly reached from i, and the so-called *backward star*, $\Delta^-(i)$, which is defined as the set of vertices j :$(i,j) \in E$, from which i is directly reachable. In this paper, only symmetric problems are referred to, and thus $\Delta(i) = \Delta^+(i) = \Delta^-(i), \forall i \in N$. The *forward* and *backward stars* will prove to be very useful for further considerations of asymmetric cases, such as the case of operating in a field with a slope area in which only one operating direction is allowed in some tracks.

$\Delta(i) \subseteq N, i \in N$ as the set of the nodes j that is directly reachable from i. We also define the function $\gamma(\cdot) : \mathcal{N} \to \{0, 1\}$ that results the parity of a number ($\gamma(i) = 1$ if i is an even number and $\gamma(i) = 0$ if i is an odd number). As proven in Appendix I, the nodes of a sub-area that are accessible from a node of the same sub-area (Fig. 3) are given by

$$\Delta(i) \cap A_k = \{j \in A_k | \gamma(i) = \gamma(j)\} \cup \{i + 1 - 2\gamma(i)\}$$

In order to provide the connectivity of the graph between the sub-areas, it is necessary to introduce the operator$\delta\langle A_k^\alpha, A_l^\beta \rangle$, where$\alpha, \beta \in \{[0], [1]\}$, $k, l \in \{1, \ldots, m\}, k \neq l$, which is given by

$$\delta\langle A_k^\alpha, A_l^\beta \rangle = \begin{cases} A_l^\beta \text{ if the } \alpha \text{ headland of area } k \text{ is connected to} \\ \text{the } \beta \text{ headland of area } l \\ \varnothing \text{ otherwise} \end{cases}$$

Using this definition, the forward star of a node $i \in A_k$ is given by

$$\Delta(i) = \{j \in A_k | \gamma(i) = \gamma(j)\} \cup \{i + 1 - 2\gamma(i)\} \cup \bigcup_{\substack{l=1 \\ l \neq k}}^{m} \delta\langle A_k^{[\gamma(i)]}, A_l^{[1]} \rangle \cup \delta\langle A_k^{[\gamma(i)]}, A_l^{[0]} \rangle$$

For an area of coverage planning, $2m\ (m-1)$ connections should be checked and inserted as an input of the problem. An example is given in Fig. 4, where there are three field areas ($m = 3$). In the specific example, 12 connections in total were checked. Given the particular configuration of the sub-fields, a machine can drive from/to sub-field 1 to/from sub-field 2 only through their lower headlands; and from/to sub-field 2 to/from subfield 3 only through their upper headlands, but it cannot drive directly from/to sub-field 1 to/from sub-field 3. Thus, the connected pairs of sets are $A_1^{[0]} - A_2^{[0]}$ and $A_2^{[1]} - A_3^{[1]}$.

There are two methods of dealing with a graph that includes missing arcs. The first method is to define the cost of each missing arc equivalent to $+\infty$ which, in practice, equals a large positive value corresponding to the cost of the connected arcs. The second method, which is preferable in the case of a large number of missing arcs, is to explicitly incorporate the *forward* and *backward stars* of the vertices in the problem's model in order to take advantage of the graph scarcity.

3.2. *Node demand*

For each node of the graph, a demand is assigned. Because each field track is represented by two graph nodes, the total demand of a track is equally split between these two nodes.

The type of demand depends on the operation type. In IMF operations, the demand of a node represents the quantity of the material (such as seed, fertiliser, and spray solution) that has to be distributed over the field area represented by the node. In the case of OMF operations, the demand of a node represents the quantity of the material that a PU has to pick up from the field area represented by the node. Finally, in NMF operations the demand of a node is inconsequential and can be defined as zero.

As previously mentioned, the characteristics of the demand can be deterministic (in NMF and IMF operations),

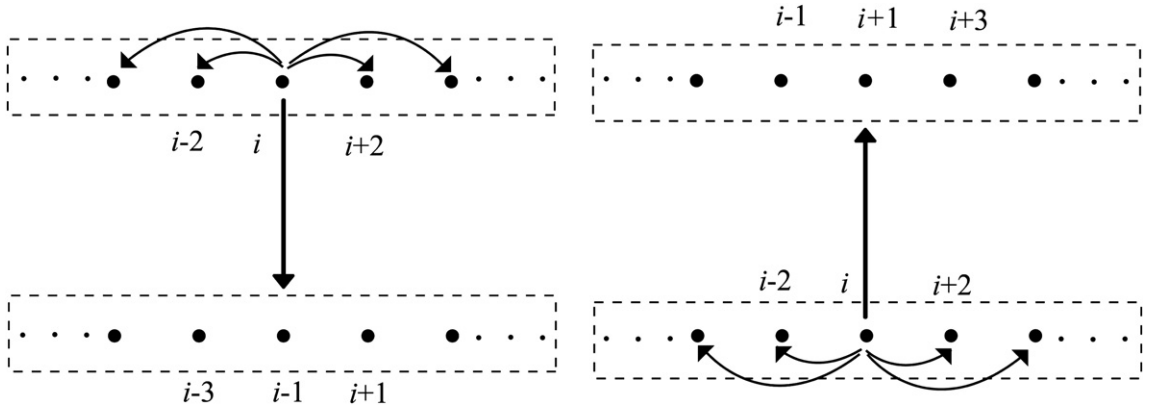


Fig. 3 – Connectivity of the nodes in the problem's graph.

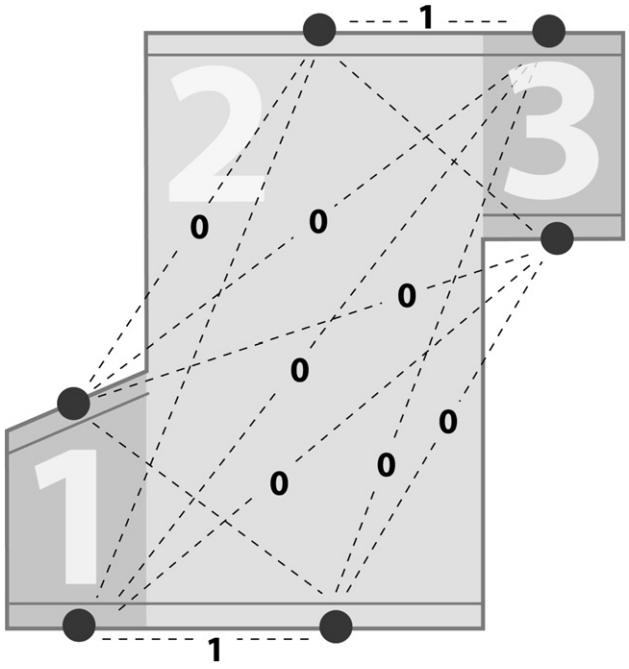


Fig. 4 – Example of field coverage planning.

stochastic (in OMF and IMF operations) and unknown (also in both OMF and IMF operations).

3.3. *Arc cost assignment*

According to the previous representation of a field area, there are three types of arcs between each pair of the graph nodes, namely: missing arcs (as described in the previous paragraph); arcs that connects pairs of nodes representing the two ends of a track; and, finally, arcs corresponding to a headland turning. In the case where two nodes represent the two ends of a track, the connection of these two nodes has to be enforced. For these nodes, the edge-traversal cost is defined as zero. The assigned cost to the arcs of the last type is determined by the non-productive time[5] the machine spends during the corresponding headland turning.

Headland turning is a function of the minimum turning radius of the machine (r_{min}), the headland clearance, the distance between the two sequential tracks, and the angle of the headland to the operating direction. It is also affected by the presence of obstacles, the soil condition and the carried implement. Fig. 5 presents four examples of headland turnings imposed by the presence of various constraints. In example (1), the distance between the two successive tracks is longer than twice the minimum turning radius of the machine. Consequently, the unit can easily perform a Π-turn. In example (2), the latter condition does not exist and the unit has to perform an Ω-turn. In addition, in example (3), the presence of an obstacle restricts the unit to performing a T-turn. Finally, example (4) refers to a dynamic case in which the unit has to seek for an alternative path in order to avoid a moving obstacle. For the generation of a headland turning path and consequently for the calculation of its length, a number of methods are available. Bochtis (2008) presented a geometrical model for calculating the minimum length of the most common types of headland turns (Π-turn, Ω-turn, T-turn and a number of their variations), as executed by an Ackerman-steering tractor. Kinematic models have also been introduced. Kise *et al.* (2002) developed a turning algorithm for a robot tractor for two types of turning paths (Ω-turn and T-turn). In Vougioukas *et al.* (2006), the presence of obstacles and headland clearance were taken into account using a two-stage motion planning algorithm. The tractor-trailer combination was investigated by Oksanen and Visala (2007) using a dynamic model approach. Further dynamic path planners for agricultural units working as multiple robots have been proposed (Vougioukas and Sigrimis, 2007). This relates to the example 4 in Fig. 5. Finally, different types of machine kinematics have been investigated in Miller *et al.* (2004).

If $\tilde{h}(i,j)$ is denoted as the time a machine needs in order to execute a headland turning from the end of the track, represented by node i, to the start of the track, represented by node j,

[5] The use of non-productive time instead of distance is preferable, due to the fact that there is a number of headland turnings where the machine has to stop and move backwards, as, for example, in a T-turn (e.g., Sørensen and Nielsen, 2005).

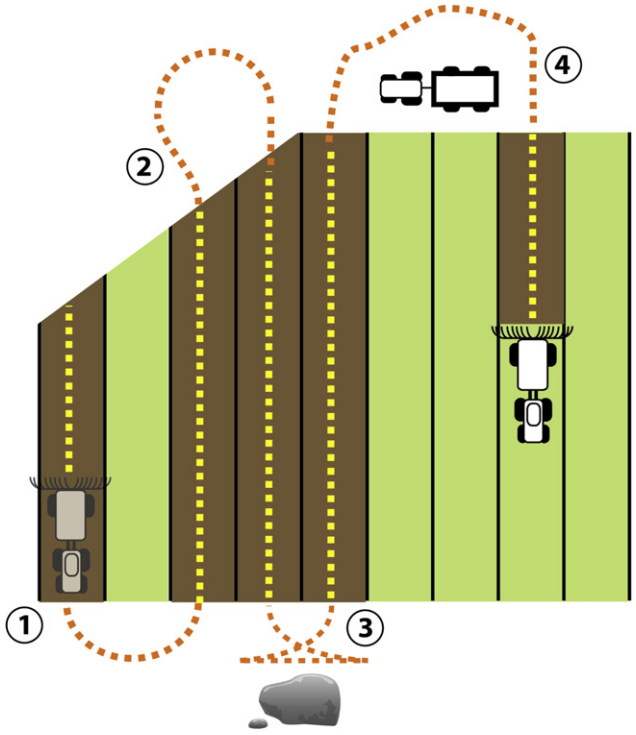


Fig. 5 – Examples of headland turnings.

the elements of the cost matrix of the corresponding VRP are given by

$$c_{ij} = \begin{cases} \tilde{h}(i,j) & j \in \Delta(i)/\{i+1-2\gamma(i)\} \\ 0 & j = i+1-2\gamma(i) \\ +\infty & j \notin \Delta(i) \end{cases}$$

3.4. *Application of vehicle routing problem instances to agricultural field operations*

In the following, starting from the simplest instance of VRP (i.e., TSP) and adding complexity to the optimisation problem, it will be shown how a series of AFOs can be solved. The cases addressed cover the majority of different types of AFOs.

3.4.1. *Single PU in an NMF operation*

$1 - \mathrm{PU}|\mathrm{NMF} \leftrightarrow \mathrm{TSP}$

The case of a single PU operating as an NMF operation can be cast as a TSP. The VRP is a generalisation of the TSP that consists of finding minimum cost closed tour for a single vehicle ($|V| = 1$) through a given set of customers, visiting each customer exactly once without the presence of capacity constraints ($C \geq \sum_{i \in N/\{0\}}$, di or equivalently, $d_i = 0, \forall i \in N$).

The solution in this case is a single circuit tour (*Hamiltonian cycle*): $R = \langle i_1, i_2, \ldots, i_{|R|} \rangle$. This sequence of nodes is transformed into a sequence of field tracks that is given by (Appendix II)

$$\left\langle \frac{i_2 \bullet \gamma(i_2) + i_3 \bullet \gamma(i_3)}{2}, \frac{i_5 \bullet \gamma(i_5) + i_5 \bullet \gamma(i_5)}{2}, \ldots, \frac{i_{|R|-2} \bullet \gamma(i_{|R|-2}) + i_{|R|-1} \bullet \gamma(i_{|R|-1})}{2} \right\rangle$$

This sequence represents the fieldwork pattern that has to be followed by the machine.

3.4.2. *Multiple-PU in NMF operations*

$m - \mathrm{PU}|\mathrm{NMF} \leftrightarrow m - \mathrm{TSP}$

In this case, a team of identical machines is operating as a NMF operation. This case can be cast as a multiple-TSP (m-TSP). The m-TSP is an extension of the TSP in which more than one route must be determined. The solution of this problem results in a fieldwork pattern for each PU p_j

$$\left\langle \frac{i_2 \bullet \gamma(i_2) + i_3 \bullet \gamma(i_3)}{2}, \frac{i_4 \bullet \gamma(i_4) + i_5 \bullet \gamma(i_5)}{2}, \ldots, \frac{i_{|R_j|-2} \bullet \gamma(i_{|R_j|-2}) + i_{|R_j|-1} \bullet \gamma(i_{|R_j|-1})}{2} \right\rangle, \; j \in P$$

3.4.3. *Single PU in a deterministic IMF operation using a single stationary FU*

$1 - \mathrm{PU}| + \mathrm{IMF}|1 - \mathrm{locFU}| \leftrightarrow \mathrm{VRP}$

This type of operation refers to a case in which there are capacity constraints to the load that a PU can to carry. Here, it is not able to complete its operation at once, so consequently, it has to leave the field and return to continue the operation one or more times. For example, in the case of spraying, the machine has to return to an out-of-field facility or to the farm's depot in order to refuel its spray tank.

This case can be cast as the classic VRP, where a "hypothetical" team of exactly $|P| = \lceil \sum_{i \in N \setminus 0} d_i / C \rceil$ identical machines is considered, and where C is the capacity of the PU. The solution will result in $|P|$ routes that are equivalent to $|P|$ tasks for the PU. Within each task, the PU has to be refilled in the FU and continue operating in the field area in the given form of the solution pattern of the current route.

3.4.4. *Multiple identical PUs in a deterministic IMF operation using a single stationary FU*

$\equiv m - \mathrm{PU}| + \mathrm{IMF}|1 - \mathrm{locFU}| \leftrightarrow \mathrm{VRP}$

It is possible to designate two different cases in this problem instance, both of which are cast as the classic VRP:

There are exactly $|P| = \lceil \sum_{i \in N \setminus 0} d_i / C \rceil$ identical PUs. In this case, a route/task is assigned to each PU.

The number of PUs exceeds $\lceil \sum_{i \in N \setminus 0} d_i / C \rceil$. In this case, a fixed cost can be associated with the use of each PU, and the additional objective of minimising the number of routes that the PUs have to use is added to the objective of minimising the total cost.

3.4.5. *Multiple non-identical PUs in a deterministic IMF operation using a single stationary FU*

$\neq m - \mathrm{PU}| + \mathrm{IMF}|1 - \mathrm{locFU}| \leftrightarrow \mathrm{MFVRP}$

The PUs that compose the team can have different capacities or different kinematics (minimum turning radius); or both, which is the general case and will be considered here. Due to the different kinematics, the turning time is different. Consequently, a different arc cost has to be determined for each machine, i.e., $c_{ij}^{k}, (i,j)\in E, k\in P$. Furthermore, the third constraint in the classic VRP case should be modified to: $\sum_{h=2}^{|R_j-1|} d_{i_h} \leq C_{p_j}$, where C_{p_k} is the capacity of the p_k PU.

This case can be cast as the mixed-fleet vehicle routing problem (MFVRP). The MFVRP is an important variant of the VRP, in which a mixed fleet of vehicles, characterised by different capacities and costs, is available for the distribution activities. MFVRP is obviously closer to practical distribution problems than the classic VRP (Baldacci *et al.*, 2007).

3.4.6. *Single PU in a stochastic OMF or IMF operation using a single stationary FU*

$1-\mathrm{PU}|(\mathrm{OMF}\vee\mathrm{IMF})|1-\mathrm{locFU}|\leftrightarrow\mathrm{VRPSD}$

In this problem, the demand at each location (yield or input such as fertiliser and seeds) is unknown at the time of planning, but it is assumed to follow a known probability distribution. The planning of such an operation can be cast as a VRP with stochastic demands (VRPSD). The VRPSD is a probabilistic variation of the classic VRP. Here, customer demands follow known probability distributions, but actual demands are only revealed when the vehicle arrives at each customer. Consequently, a route designation may fail if customer demand exceeds the current vehicle capacity, and a recourse action, such as sending the vehicle back to the depot and forth to the customer, must be taken at extra cost. In the case of a harvester that travels to a fixed unloading location in order to unload its temporary grain hopper, the operator has to choose, according to his estimation of the relationship between the expected yield and the length of the track, whether to proceed to the next track or to go on to the unloading location. Sometimes choosing to restock is best, even if the grain hopper of the harvester is not full, or if its capacity is larger than the expected demand of the next scheduled track according to the fieldwork pattern. The analogous action in the case of vehicle routing is called '*preventive restocking*'. The goal of preventive restocking is to avoid the undesirable situation in which the grain hopper of a harvester becomes full before the end of the field track, and the harvester has to manoeuvre in order to both leave the field and to resume the operation from the last working position. It is important that the operator chooses appropriate strategies in order to cope with such situations.

In this type of operation the demands of the field tracks (yield or application needs) are independently distributed within known distributions. A number of methods have been developed to deal with the VRPSD of a single vehicle, both in cases which do not consider the possibility of doing preventive restocking (Dror, 1993) as well as, to those that do (Yang *et al.*, 2000; Bianchi *et al.*, 2004). An optimal choice for the case of preventive restocking in the case of harvesting, for example, is of the threshold type. That is, given the *a priori* field work pattern, for each track $t\in T$, there is a load threshold ϑ_t such that if the residual grain hopper capacity load after serving t is greater than or equal to ϑ_t, then it is better to proceed to the next track; otherwise it is better to travel to the FU to unload the grain hopper.

3.4.7. *Multiple PUs in OMF or stochastic IMF operation using one or multiple stationary FU*

$\mathrm{m}-\mathrm{PU}|\sim(\mathrm{OMF}\vee\mathrm{IMF})|(1\vee\mathrm{m})-\mathrm{locFU}|\leftrightarrow\mathrm{VRPSD}$

Examples for multiple vehicles and multiple depots have been developed from those of a single vehicle using a single depot (Teodorovic and Pavkovic, 1992; Gendreau *et al.*, 1996). These instances are applicable to field operations involving multiple machines and FUs.

3.4.8. *Summary*

Based on the identified instances of the VRP as related to AFOs, the types of operation are summarised in Table 2.

Table 2 – VRP instances and their corresponded field operation types.

Operation type	VRP instance	Examples
1-PU | NMF	TSP	Arable farming: tillage (seedbed preparation, cultivation, *etc.*), mowing, grass/hay conditioning, grass/hay baling, windrowing. Orchard Farming: mechanical weeding, inter-row mowing.
m-PU | NMF	m-TSP	The NMF operation types executed by a team of identical machines (e.g., field robots)
1-PU | + IMF | 1-staFU	VRP	Arable farming: Spraying (uniform application or map-based variable rate application), seeding, fertilizing (uniform application or variable rate application), Orchard farming: spraying using mist blower for pest control, spraying using nozzle sprayers for weed control.
≡m-PU | + IMF | 1-staFU	VRP	The VRP and IMF operation types executed by a team of identical machines (e.g., field robots)
≠m-PU | + IMF | 1-staFU	MFVRP	The IMF operation types executed by a team of non-identical machines (e.g., field robots)
1-PU | ∼IMF | 1-staFU	VRPSD	An application unit in a sensor-based variable rate precision spraying with some *a priory* information (e.g., satellite image).
m-PU | ∼IMF | (1∨ m)-staFU	VRPSD	The IMF operation types executed by a team of identical machines (e.g., field robots)
1-PU | ∼OMF| 1-staFU	VRPSD	A harvester (e.g., grain, cotton) that unloads its bin at a predetermined out-of-field location.
m-PU |∼OMF | (1∨ m)-staFU	VRPSD	The OMF operation types executed by a team of identical machines (*e.g* field robots)

4. Discussion

The objective of this paper has been to describe how, using an abstraction of to represent the field area, a number of agricultural fleet management cases can be cast as various instances of the VRP. The inclusion of case studies or experimental results is beyond the scope of this paper as each of the presented operation types constitutes further individual research efforts.

One constraint of applying of the VRP in AFOs executed by non-identical agricultural machines is that the working width should be the same for all machines. In the case of homogenous PUs with different working widths operating in an area coverage operation, a graph representing the problem cannot be determined. This is due to the fact that different working widths result in different numbers of field tracks and, consequently, in different numbers of graph nodes. The proposed solution for this case is the assignment of field parts to each PU based on their theoretical field efficiency. Subsequently, each part corresponds to a single machine VRP instance and that can be solved accordingly.

It should be mentioned that the advanced technologies that are implemented, both now and in the future, make it feasible for mobile agricultural machinery to be subjected to optimal planning and control measures in relation to the perception of a human operator including the optimal paths generated by the implementation of VRP-solving algorithms. The next generation of auto-steering and navigation add-on systems is expected to incorporate capabilities, such as automated headland turning, algorithmic determination of track sequences, and off-road navigation.

The proven casting of VRP instances for the planning of AFOs provides an insight into the complexities issues of such operations. Since an AFO planning problem can be cast as an instance of VRP, both problems are of similar complexity. As mentioned earlier, VRP is a generalisation of one of the most famous combinatorial optimisation problems, the TSP. The TSP is an NP-hard problem[6] (Papadimitriou and Steiglitz, 1998). This means that it is not always possible to find optimal solutions for problems of realistic size in a reasonable amount of computation time. Even for small fleet sizes and a moderate number of transportation requests, the planning task is of high complexity. In a problem with n field tracks, for example, the implementation of a direct solution consisting of the systematic enumeration of all possible ordered combinations, in order to find the optimal sequence of tracks in terms of non-working travelled distance in headlands, would demand almost $n!$ computational steps. This makes the solution for as few as 50 field tracks impractical, since an enormous length of time would be necessary to solve this scale of problem even using the most powerful modern computers (Bochtis, 2008). The adoption of solving methods and the implementation of powerful algorithms developed for VRP solutions can provide an efficient way of dealing with the complexity inhered in AFO) planning problems.

[6] Non-deterministic polynomial-time derived from computational complexity theory.

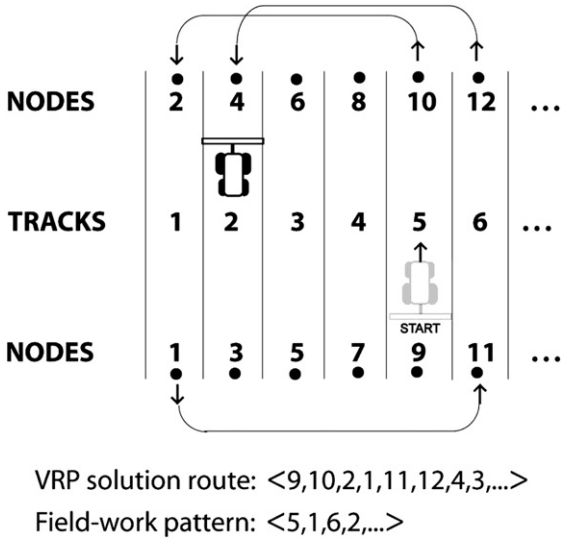


Fig. 6 – VRP route transformation to track sequence (field-work pattern).

Also, agricultural fleet management must face the reversal of current trends that involve increasingly heavy machinery systems and look towards the development and application of teams of smaller, lighter, and more intelligent field agents. It is the authors' belief that, in light of emerging field robots in agriculture, the implementation of VRP techniques for effective mission and route planning is a crucial and determining prerequisite for their use.

5. Conclusions

In this paper, a theoretical approach to the representation of the planning for AFOs executed by primary machines as instances of the VRP has been presented. By applying an abstractive representation, it has been shown that AFO executed by agricultural machines (manned or unmanned) are equivalent to the traversal of weighted graphs. Additionally, depending on the type of operation, they can be cast as a VRP instances and, consequently, can be solved using developed methods for the solution of these instances. The abstraction involved the representation of field tracks as the ''customers'' in the VRP.

By casting AFO problems in the framework of well-known VRP instances, over fifty years of research and implementation experience in ''*one of the great success stories of operational research*'' as Toth and Vigo (2002) put it, can provide significantly more efficient management of conventional machinery systems at a specified level of automation. It will also improve the mission and route planning aspects for the high-level control of emerging field robots.

Appendices

Appendix I: Graph connectivity

If node i belongs to the upper headland ($i \in A_k^{[1]}$), the other node of the track is node $i-1$. If node i belongs to the lower

headland ($i \in A_k^{|0|}$), the other node of the track is given by $i+1$ (Fig. 3). By adding the latter, it is possible to write the "complement" of a node i as given by: $j = \begin{cases} i+1 \text{ if } \gamma(i)=0 \\ i-1 \text{ if } \gamma(i)=1 \end{cases} \Rightarrow j = i+1-2\gamma(i)$. Note that, in order to enforce the connection of the two nodes representing the two endings of a track, it is necessary to define $c_{i,i+1-2\gamma(i)} = 0$. Except for the complement of a node, all nodes belong to the same headland where i belongs. In other words, the nodes with the same parity as i should be included in set $\Delta(i)$.

Summarising the previous, for a node$i \in A_k$, the connected nodes of set A_k constitute the set: $\Delta(i) \cap A_k = \{j \in A_k | \gamma(i) = \gamma(j)\} \cup \{i+1-2\gamma(i)\}$

Appendix II: Node sequence to track sequence transformation

The VRP solution results in a sequence of nodes (representing a route) for each vehicle: $R_j = \langle i_1, i_2, \ldots, i_{|R_j|} \rangle, j \in V$. These sequences must be transformed to track sequences according to how each machine should visit its assigned tracks. Each track is represented by a pair of nodes. One track is of parity 1 (even number) and the other of parity 0 (odd number). Considering the representation of the tracks, when the larger of the two numbers is the even ($\gamma(\max[m,n]) = 1$), and that the track number equal to this number divided by 2, the node sequence is transformed to (Fig. 6)

$$\left\langle \frac{i_2 \cdot \gamma(i_2) + i_3 \cdot \gamma(i_3)}{2}, \frac{i_5 \cdot \gamma(i_5) + i_5 \cdot \gamma(i_5)}{2}, \ldots, \frac{i_{|R_j|-2} \cdot \gamma\left(i_{|R_j|-2}\right) + i_{|R_j|-1} \cdot \gamma\left(i_{|R_j|-1}\right)}{2} \right\rangle$$

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